Research protocol

Adoption and evaluation of the Doha agreement classification system of groin pain in athletes (part 2): a worldwide survey

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1. Introduction

Hip- and groin injuries are commonly incurred by athletes participating in high intensity sports involving change of direction and kicking, such as ice hockey or football.[1] In football, groin injuries account for 7-13% off all injuries when time-loss or medical attention definitions of injury are used.[2] If a non-time loss definition of injury is used, as many as 50% of players on a football team may experience hip-and/or groin pain during a single season.[3,4]

The diagnostic terminology used by clinicians and researchers to classify groin injuries varies substantially.[5,6] A previous Delphi study reported up to 22 different terms used by experts to classify two “real life” groin injuries presented by way of clinical vignettes.[6] Similarly, a systematic review on groin injury management identified 33 different diagnoses under the umbrella term “groin injuries” across the 72 included studies.[5] A number of consensus statements have been published on diagnostic terminology for groin pain, due to the absence of a definitive or universally agreed upon reference standard.[7–9].

In 2014, the British Hernia Society’s position statement focused on the inguinal region. They proposed that the term “inguinal disruption” should be used instead of the terms “sportsman’s hernia/groin”.[8] They defined inguinal disruption as pain located around the pubic tubercle, without the presence of clear pathology (such as an inguinal hernia). Additionally, it was suggested that 3 out of 5 possible clinical signs should be present for a positive diagnosis: 1. Pinpoint tenderness over the pubic tubercle at the point of insertion of the conjoint tendon; 2. Palpable tenderness over the deep inguinal ring; 3. Pain and/or dilation of the external ring with no obvious hernia evident; 4. Pain at the origin of the adductor longus tendon; and 5. Dull, diffused pain in the groin, often radiating to the perineum and inner thigh or across the midline.

In 2015, the Doha agreement meeting resulted in a clinical classification system based primarily on clinical examination findings as follows: (1) defined clinical entities for groin pain: adductor-, inguinal-, iliopsoas-, and pubic-related groin pain, (2) hip-related groin pain, and (3) other causes of groin pain.[7] Each defined clinical entity for groin pain has specific diagnostic criteria, generally based on localized tenderness and pain on resistance testing and described in detail in the paper.[7]

In 2016, the Italian consensus meeting reported the umbrella term “groin pain syndrome”, listing 63 potential causes of groin pain.[9] In this consensus statement, diagnosis and treatment decision is suggested to be based on history, clinical examination and imaging findings.

Since the publication of these consensus statements it is unknown if clinicians who assess athletes with groin pain use any of them in their daily clinical practice. Therefore, the aim of this worldwide survey is to evaluate to what extent the recommendations on diagnostic terminology of these consensus statements have been adopted by clinicians who assess athletes with groin pain.

2. Study objective

To evaluate to what extent the recommendations on diagnostic terminology of available consensus statements have been adopted by clinicians who assess athletes with groin pain.

3. Methods

An e-survey was developed using the Checklist for Reporting Results of Internet E-Surveys (CHERRIES).[10]

3.1 Research sample

Experts will be recruited by contacting all 49 FIFA Medical Centres of Excellence (FMCE) and 11 International Olympic Committee (IOC) research centres.

The FMCEs were selected, since groin injuries are prevalent in football [1–4], and these centres cover a wide geographical region. The centers are located in 32 countries in Asia, Africa, Europe, Australia, North America and South America, and they must demonstrate their leadership in football medicine through a strict selection process. Although football medicine has a prominent role in these centres, they also see athletes performing other sports. *(*https://www.fifa.com/what-we-do/medical/)

The 11 recognized IOC centres are tasked with researching, developing and implementing effective preventive and treatment methods for sports-related injuries and illnesses. All IOC research centres are located across 5 continents, and most research centres consist of different institutions. (https://www.olympic.org/news/ioc-recognises-11-research-centres-worldwide-for-prevention-of-injury-and-protection-of-athlete-health)

Two centres are recognized as both a FMCE and IOC-centre, thus in total **58 centres will be contacted**.

In- and exclusion criteria for survey respondents

Each centre will be contacted by email and asked to provide the email addresses of 3 (para)medical staff members from their centre who meet the criteria below:

Inclusion criteria:

- A health professional (physiotherapist, orthopaedic surgeon, general surgeon, sport medicine physician or family doctor) who assessed at least 1 athlete with groin pain every month (on average, during the past year)
- Able to understand and provide answers in English language.

Exclusion criteria:

- Experts/authors from the British Hernia Society's 2014 Position Statement[8], the 2015 Doha agreement classification[7], or the Italian groin pain syndrome 2016 consensus[9]
- Respondent answers that he/she assessed <1 athlete with sports-related groin pain on average each month during the past year (demographical questions)

Each center will be encouraged to select clinicians from different sexes and various backgrounds (e.g. 1 male physiotherapist, 1 female sports medicine physician and 1 male orthopaedic surgeon). However, if a centre for example only has 3 male physiotherapists who manage athletes with groin pain, this will be acceptable. The centres will also be told that the years/level of experience of the clinicians may vary.

Participation of selected clinicians is voluntary, and there will be no incentives offered to respondents (e.g., monetary, prizes, or non-monetary incentives).

Each centre will have the chance to provide the email addresses of 3 clinicians within a 3-week time frame after the initial email. An email reminder will be sent to non-responding centres after 1.5 week.

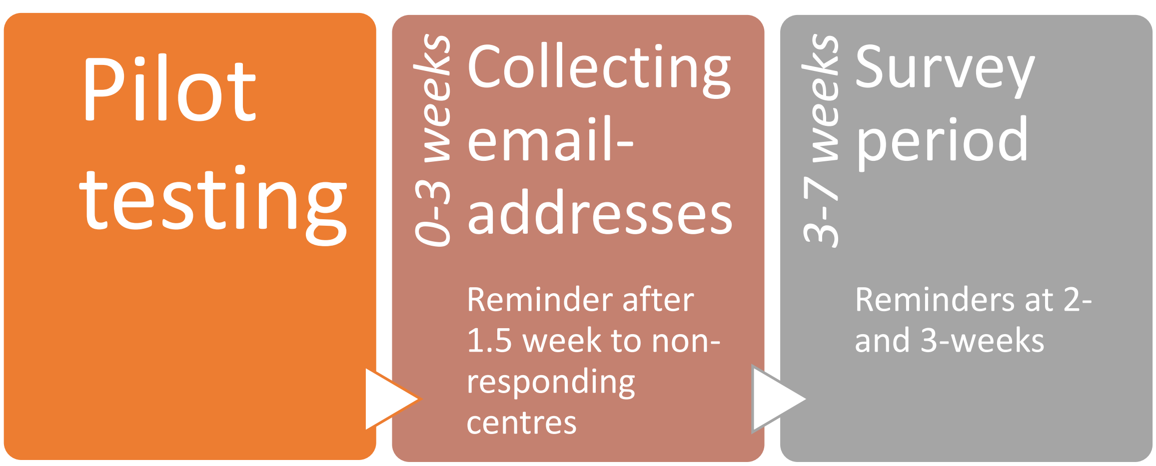
3.2 Procedures

Informed consent and ethics
Participants will be informed about the expected length of the survey, study purpose, involved investigators and data management. The study purpose of the survey is defined to participants as: “to determine which diagnostic terminology clinicians use in their daily practice when clinically assessing athletes with groin pain”. An (electronic) informed consent must be given before proceeding to the survey content. An ethics exemption for this study was obtained from the Institutional Review Board of the Amsterdam Medical Centre in the Netherlands.

Development
All authors were involved in developing the content for this survey. Google Forms will be used as the survey method. All authors were asked to test the final version of the survey in Google Forms, and provide feedback.

Duration and Data management
The survey will be open for responses for a duration of 4 weeks. Email reminders will be sent at 2- and 3-weeks after the initial email. A unique code will be assigned to each respondent. Responses will be housed in the Google Forms database, with the research coordinator being the only person with access to this database. Once all respondents have completed the survey, data will be exported from the Google Forms database to a password-protected computer and a secure (password-protected) external hard drive. This anonymised data will be stored indefinitely.

Flow chart of study procedures



Survey content (Appendix)
This survey contains 9 sections, including the electronic informed consent. A “back” and “next” button will be available, giving participants the option to change answers while completing the survey. Respondents are free to decline an answer if they want. However, only the questions involving

in/exclusion criteria and the main research question must be answered in order to have a valid survey response. Once all answers have been submitted, it will not be possible to amend any answers.

The 5 main sub-headers of this survey are:

1. **Introduction/aim and informed consent (appendix section 1)**
2. **Demographical information (appendix section 2)**
3. **Diagnostic classification system (appendix sections 3-6)**

Answer options for the preferred classification system of the respondent will be randomized using the randomization option of Google Forms. Adaptive questioning (i.e. a follow-up question will be displayed to clarify a specific answer) will be used if a respondent answers that he/she does not use any of the three mentioned classification systems or uses a combination of classification systems.

4. **Diagnostic terms for 2 clinical cases (appendix section 7-8)**

The 2 clinical cases are identical to case 1 and 2 of “Adoption and evaluation of the Doha agreement classification system of groin pain in athletes (part 1): a Delphi study”. The third case from part 1 was excluded, specifically to limit the time required to complete the survey. For each case, the injury history and clinical examination findings will be extensively described, similarly to the Delphi study (part 1). To ensure understanding of all descriptions and described tests for non-native English-speaking respondents, only minor language editing/changes were allowed to provide more clarity.

Each respondent will be asked to provide 1 or 2 terms to describe his/her suspected clinical diagnosis.

5. **General comments (appendix section 9)**

Any general comments on this topic or this survey can be shared here in an open text field.

Preventing multiple entries from the same individual
After submitting the form, the respondent will see a confirmation message that his/her response has been recorded and will be requested to only respond once.

Terms provided for the 2 clinical cases
All provided terms will be recorded separately. For example: “adductor tendinopathy” and “adductor tendinitis” will be recorded separately. In case of uncertainty/unclarity around a described term, it will be assessed and decided by consensus (WH, AS, ZV) if the term was a:

- Typographical error: it will be changed appropriately
- Unclear or very a-specific (such as “groin pain”, “adductor”): the term will be excluded.

If a respondent provided more than 2 terms, term 3 (and above) will be excluded.

Protocol registration
The study protocol will be registered on the website “https://protocols.io”.

4. Statistics

<u>Study objective</u>
To evaluate to what extent the recommendations on diagnostic terminology of available consensus statements for groin pain in athletes have been adopted by clinicians who assess athletes with groin pain.

The statistical analysis solely comprises descriptive statistics. Numbers (%) will be reported for categorical data. Normally distributed continuous data will be reported as mean (standard deviation). Non-normally distributed continuous data will be reported as median (interquartile range). Response rate will be calculated by dividing the number of responses by the number of invitations sent out.

Provided terms for the diagnosis of the 2 clinical cases
The number of each given term/diagnosis will be divided by the total number of terms/diagnosis provided for each case (max. 2 terms per respondent).

References

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Appendix

Section 1: Introduction and aims / informed consent

Thank you for considering taking part in this survey on groin pain in athletes. This survey takes about 10-15 minutes. Your response will be anonymous.

The aim of this survey is to determine which terminology clinicians use in their daily practice when clinically assessing athletes with groin pain.

You will be asked a few general questions and then presented with two clinical cases and some associated questions. You will be asked a few general questions and then presented with two clinical cases and some associated questions. Some answers may not require a follow-up question (i.e. you may see that you move from page 3 to 7).

Your answers are important to the field of sports and exercise medicine and will contribute to an enhanced understanding of the specific terminology that clinicians use when assessing athletes with groin pain.

Please check the in- and exclusion criteria for participation on this survey:

I confirm that I am:

1. A health professional (physiotherapist, orthopaedic surgeon, general surgeon, sport medicine physician, radiologist or family doctor) who assessed at least 1 athlete with groin pain every month (on average, during the past year)
2. Able to understand and provide answers in English language

I confirm that I have not participated in the:

1. The British Hernia Society's position statement (2014)
2. The Doha agreement classification (2015)
3. The Italian groin pain syndrome consensus (2016)

Informed consent

- I confirm that I have read and understand the aim of the survey
- I understand that my participation is voluntary
- I understand that, should I not wish to answer any particular question or questions, I am free to decline
- I give permission for my anonymised responses to be used by members of the research team
- I understand that my anonymised responses will be aggregated with those of other respondents and will be used by the research team in future scientific presentations and academic publications

If you click NO and choose not to participate in the study, you will be disqualified from continuing. If you click YES and consent to participate in the study, you will automatically continue to participate in the study.

Informed Consent

Q: Do you fully consent to participate in this survey on the diagnostic terminology on groin pain in athletes used by clinicians?

A: - Yes (moves to section 2)
- No (stops the survey)

Data storage
A unique code will be assigned to each respondent. Responses will be housed in the Google Forms database, with the research coordinator being the only person with access to this database. Once all respondents have completed the survey, data will be exported from the Google Forms database to a password-protected computer and a secure (password-protected) external hard drive. This anonymised data will be stored indefinitely.

Section 2: Demographical information

Q: Gender:

A: - Male
- Female

Q: Age:

A: Free text

Q: In which country are you working?

A: Free text

Q: Profession:

A: - Sports Physician
- Physiotherapist
- General surgeon
- Orthopaedic surgeon
- Radiologist
- Family medicine physician
- Other (free text)

Q: *Number of years working since qualification?*

A: Free text

Q: *How many athletes with groin pain have you clinically assessed per month? (on average during the past year)*

A: free text

Section 3: Diagnostic classification system

Q: *Which diagnostic terminology or clinical classification system(s) are you using when clinically assessing athletes with groin pain? (order is randomized)*

A:
- The Doha agreement classification (2015) (moves to section 6)
- The British Hernia Society's position statement (2014) (moves to section 6)
- The Italian groin pain syndrome consensus (2016) (moves to section 6)
- I use a combination of classification systems (moves to section 5)
- I don't use a classification system (moves to section 4)
- Other (free text) (moves to section 4)

Section 4: Diagnostic classification system

Q: What is your reason for not using any of the three mentioned classification systems?

A:
- I don't know these classification systems (moves to section 7)
- I know the classification systems, but prefer a different approach (moves to section 5)
- Other (free text) (moves to section 5)

Section 5: Diagnostic classification system

Q: Can you elaborate on you reason for not using any of the three mentioned classification systems?

A: Free text

Q: Can you elaborate on your preferred classification system/diagnostic terminology for athletes with groin pain?

A: Free text

(moves to section 7)

Section 6: Diagnostic classification system

Q: Which combination of classification systems are you using?

A:	- The British Hernia Society's position statement (2014)	(moves to section 7)
	- The Doha agreement classification (2015)	(moves to section 7)
	- The Italian groin pain syndrome consensus (2016)	(moves to section 7)
	- Other (free text)	(moves to section 7)

Section 7: Clinical case 1 (identical to case 1 in Delphi study)

History:

- 30-year-old male CrossFit/fitness athlete with left-sided groin pain
- Trains 6 days a week: mixture of high intensity weightlifting, running and body weight exercises
- No past history of groin pain

Past medical history:

- Tonsillectomy as child
- Medication: none
- Allergies: none
- Social: works in ICT – desk job, non-smoker, less than 5 units alcohol per week

Injury history:

- 6 months ago during training did lots of explosive sit ups during one week from full lumbar extension
- No acute moment or sensation of popping/tearing
- The pain gradually built up during the sit ups
- After the heavy training week he had pain in the left lower abdominal region
- Pain in the same area on coughing and defecation without radiation into the testicles or perineum
- There was no visible swelling
- Pain at night on turning over in bed
- Initial pain with activities of daily living and at night settled after the first week
- The pain persisted when he tried to resume sporting activities such as running or lifting weights in the gym
- Due to the persistent pain in left lower abdominal area on activity he visited his family doctor
- On clinical examination the family doctor could not feel an inguinal hernia
- He was advised to rest for a month and then start doing sports again
- Since this time he has rested each time for 4 weeks, but every time on restarting running or lifting weights pain is felt in the left lower abdominal region

Current symptoms:

- The pain is a dull, burning type pain in the left lower abdominal area
- It is worsened by activity: when lifting weights, or running faster than slow jogging
- No pain present at rest or during activities of daily living
- No pain anymore during urination or defecation
- There is no altered sensation of the skin
- He has adapted his sports and now does gentle jogging every other day for 30 minutes which is pain

free
- He lifts light weights and swims to maintain his fitness

Clinical examination:

General:
- Healthy athletic male
- Mild varus alignment both legs
- Lumbar spine – normal pain free range of motion
- Horizontal pelvis

Inspection groin region:
- No swelling, bruising, scars

Hip:
- Normal pain free flexion
- 20 degrees internal rotation of both hips - pain free
- 50 degrees external rotation of both hips - pain free
- No pain on FABER or FADIR-tests of both hips

Palpation:
- Recognisable injury pain on palpation of the left external inguinal ring superficially
- Recognisable injury pain on palpation of the left inguinal canal on scrotal invagination
- No apparent bulging of the posterior wall on Valsalva
- No palpable inguinal hernia
- No pain on palpation of adductors, iliopsoas, distal rectus abdominus, pubic bone or other structures in the groin region
- Normal sensation in the groin region

Resistance testing (isometric tests - all tests in supine position):
- Hip adduction with 45° and 90° hip flexion - no pain felt at adductor insertion, good strength
- Hip adduction with 0° hip flexion - recognisable injury pain felt in left inguinal canal, good strength
- Hip flexion with 0° and 90° hip flexion - no pain, good strength.
- Hip abduction with 0° and 45° hip flexion - no pain, good strength
- Abdominal sit up 45° hip flexion – recognisable injury pain felt in left inguinal canal, good strength
- Oblique sit ups 45° hip flexion – recognisable injury pain felt in left inguinal canal, good strength

Stretch tests:
- Symmetrical length of adductors with no pain on stretching
- Symmetrical length on testing hip flexors and no pain on stretching

All other not mentioned clinical examination test can be considered normal.

Q: *Please share the terms you would use for the most likely diagnosis or diagnoses of these patients in your practice. You can enter 1 or 2 terms below that you would use for your clinical diagnosis.*

A: Diagnosis 1: Free text
Diagnosis 2 (optional): Free text

## Section 8:	Clinical case 2 (identical to case 2 in Delphi study)

History:

- 29 year-old male top amateur football player with right sided groin pain
- Trains 3 times a week and plays a match on the weekend
- Right leg dominant
- He has previously experienced several episodes of groin pain on starting football after the summer break, normally settled within a month

Past medical history:
- Fracture left foot 3 years ago – no residual complaints
- Ankle sprain left side 5 years ago – no residual complaints
- Medication: none
- Allergies: none
- Social: works business - desk job, non-smoker, no alcohol

Injury history:
- Since the start of the season 4 months ago a gradual onset and progressive pain at the right proximal adductor insertion
- Initially able to train and play
- Now due to worsening pain unable to sprint or kick with power with the right leg
- No pain on urination, defecation or Valsalva
- He has tried resting 7-10 days several times, but the pain recurs when he returns to team training again
- No pain during activities of daily living
- He went to see his family doctor who referred him to the hospital

Current symptoms:
- The pain is a dull pain at the proximal adductor insertion but becomes stabbing when he sprints or shoots the football with his right leg
- After sprinting, the pain can radiate down the inside of the medial thigh for several hours after training
- He has no pain in the inguinal canal region
- No pain felt centrally over the pubic symphysis
- He has now stopped playing football for the last 3 weeks and currently only walks his dog 30 minutes a day for exercise

Clinical examination

General:
- Healthy athletic male
- Mild varus alignment both legs
- Lumbar spine – normal pain free range of motion
- Horizontal pelvis

Inspection groin region:
- No swelling, bruising, scars

Hip:
- Normal pain free flexion on both sides
- 10 degrees internal and 25 degrees external rotation bilaterally without pain
- No pain on FABER or FADIR tests on both hips

Palpation:
- Recognisable injury pain on the right proximal adductor longus insertion on the pubic bone
- Inguinal canals pain free during scrotal invagination
- No apparent bulging of the posterior wall on Valsalva and no palpable inguinal hernia
- No pain on the distal rectus abdominus, pubic bone or other structures in the groin region
- Normal sensation in the groin region

Resistance testing:
- Hip adduction with 0°, 45° and 90° hip flexion - recognisable injury pain at the right proximal adductor longus insertion, good strength
- Hip flexion with 0° and 90° hip flexion - no pain, good strength
- Hip abduction with 0° and 45° hip flexion - no pain, good strength
- Abdominal sit ups (hips flexed 45°) straight and oblique - pain free, good strength

Stretch tests:
- Symmetrical length of adductors with no pain on stretching
- Symmetrical length on testing hip flexors and no pain on stretching

All other not mentioned clinical examination test can be considered normal.

Q: *Please share the terms you would use for the most likely diagnosis or diagnoses of these patients in your practice. You can enter 1 or 2 terms below that you would use for your clinical diagnosis.*

A: Diagnosis 1: Free text
Diagnosis 2 (optional): Free text

Section 8: General comments

Q: Please share here if you have any additional thoughts on the topic of diagnosis and classification of groin pain or this survey in general.

A: Free text

Confirmation message

Thank you for your time!

Your response has been recorded. Please send your response only once.